

Reusable Modules

Reusable Modules ermöglichen die modulare Wiederverwendung von Testschritten innerhalb des Agate Studios. Wenn sich bestimmte Testsequenzen (wie z. B. Login-Vorgänge oder Standard-Validierungen) in mehreren Testfällen wiederholen, können diese in separate, eigenständige YAML-Dateien ausgelagert und als **Reusable Test Steps** deklariert werden. Der Aufruf erfolgt innerhalb des Haupttests über den Schritttyp `type: CALL`.

1. Reusable Modules – Kompakte System-Referenz für KI-Modelle

```
1 | YAML
2
3
4 # 1. Syntax im aufrufenden Testfall (Main Suite)
5 - type: CALL
6   command: "reusable.demo_reusable" # Pfad-Identifikator zur YAML-Datei
7   condition: "..." # Optional: '{B[var]}' == 'val' (Einfache Anführ
8   parameters: # Kontext-Spezifische Übergabewerte
9     verbose: "true|false" # Optional: Aktiviert/Deaktiviert da
10    command: "..." # Runtime-Parameter (erreichbar über {R[...]})
11
12 # 2. Syntax innerhalb des Reusable Modules (./reusable/demo_reusable.ya
13 # Dieses File enthält eine flache Liste von Standard-Schritten (CMD, RE
14 steps:
15 - type: CMD
16   op: EXEC
17   command: "{B[command]}" # Liest aus dem globalen B-Buffer (T
18   response: resp1
19
20 - type: CMD
21   op: EXEC
22   command: "{R[command]}" # Liest aus dem lokalen R-Buffer (Üb
23   response: resp1
24
```

2. Funktionsweise & Kernkonzepte

2.1. Datei-Auflösung und Pfad-Struktur

Der Typ `CALL` definiert einen wiederverwendbaren Testschritt. Der Parameter `command` dient als Identifikator für die zu ladende Datei.

- Der Engine übersetzt die Punkt-Notation (`reusable.demo_reusable`) in einen relativen Dateipfad basierend auf dem Verzeichnis der aktuellen Test-Suite.
- Aus `reusable.demo_reusable` wird somit die Datei `./reusable/demo_reusable.yaml` geladen und deren Schrittliste (`steps:`) sequentiell zur Laufzeit ausgeführt.

2.2. Das Variablen-Konzept: B-Buffer vs. R-Buffer

Für Reusable Modules gelten alle Regeln und Engine-Typen des Standard-Frameworks. Der entscheidende Unterschied liegt in der Verfügbarkeit von zwei getrennten Variablen-Speichern zur präzisen Steuerung von dynamischen Inhalten:

- **Der B-Buffer ({B[variable_name]})**: Dies ist der standardmäßige, globale Variablen-Speicher des Testfalls (unter `variables:` im Haupttest definiert) oder ein zur Laufzeit im Kontext veränderter Buffer. Reusable Modules haben vollen Lese- und Schreibzugriff auf diesen Kontext.
- **Der R-Buffer ({R[variable_name]})**: Dies ist ein **speziell für Reusable Modules** eingeführter, lokaler Speicher. Er greift direkt auf die Werte zu, die im `parameters`-Block des `CALL`-Schritts übergeben wurden.

Vergleichs-Beispiel zur Veranschaulichung:

Wenn ein Haupttest die Variable `command: "java -version"` im globalen Kontext hält und das Reusable Module mit dem Parameter `command: "mvn --version"` aufruft, gilt innerhalb des Moduls:

- `{B[command]}` wird zu `"java -version"` aufgelöst.
- `{R[command]}` wird zu `"mvn --version"` aufgelöst.

2.3. Spezifisches Logging-Verhalten (Verbose-Steuerung)

Da Reusable Modules in der Praxis meist gut erprobte, stabile Standard-Komponenten sind (z. B. ein technischer Login), sollen sie die Test-Logs nicht unnötig überladen.

- **Default-Verhalten (verbose: "false")**: Die Ausführung der einzelnen Schritte innerhalb des Reusable Modules wird standardmäßig **unterdrückt** und nicht im DSL-Log ausgegeben.
- **Aktiviertes Logging (verbose: "true")**: Wird im `parameters`-Block explizit `verbose: "true"` übergeben, schaltet der Engine das detaillierte Logging für alle im Modul enthaltenen Schritte ein.

3. Parameter-Referenz für den CALL-Schritt

Parameter	Beschreibung	Erforderlich	Standardwert
type	Muss exakt auf <code>CALL</code> gesetzt werden	Ja	-

command	Der logische Pfad zur YAML-Datei (z. B. <code>ordner.dateiname</code>)	Ja	-
condition	<i>Optional:</i> Bedingung für die Ausführung. Der Schritt wird übersprungen (<code>SKIPPED</code>), wenn die Bedingung <code>false</code> ergibt.	Nein	
parameters	Map von Schlüssel-Wert-Paaren, die als <code>R-Buffer</code> übergeben werden	Nein	-
parameters.verbose	Steuert die Log-Ausgabe des Moduls (<code>"true"</code> oder <code>"false"</code>)	Nein	<code>"false"</code>

Praxis-Tutorial: Umfassende Test-Suite mit Reusable Modules

Das folgende Beispiel zeigt eine vollständige Test-Suite (`test_suite.yaml`), die verschiedene Facetten des `CALL` -Mechanismus demonstriert, gefolgt von der Struktur des aufgerufenen Moduls.

1. Das Reusable Module: `[Suite-Verzeichnis]/reusable/demo_reusable.yaml`

```
1 | YAML
```

```
1 # This file contains steps that can be called by any TestCase
2 steps:
3     # 1. Execute and store full response based on B-Variable
```

```

4      - type: CMD
5        op: EXEC
6        command: "{B[command]}"
7        response: resp1
8
9      # 2. Assert Exit Code
10     - type: CMD
11       op: ASSERT
12       response: resp1
13       action: EXITCODE
14       expected: 0
15
16     # 3. Execute and store full response based on R-Variable
17     - type: CMD
18       op: EXEC
19       command: "{R[command]}"
20       response: resp1
21
22     # 4. Assert Exit Code
23     - type: CMD
24       op: ASSERT
25       response: resp1
26       action: EXITCODE
27       expected: 0
28
29     # 5. Execute and store full response based on R-Variable
30     - type: CMD
31       op: EXEC
32       command: "{B[command]}"
33       response: resp1
34
35     # 6. Assert Exit Code
36     - type: CMD
37       op: ASSERT
38       response: resp1
39       action: EXITCODE
40       expected: 0
41
42     # 7. Execute and store full response based on B-Variable
43     - type: CMD
44       op: EXEC
45       command: "{B[command2]}"
46       response: resp2
47
48     # 8. Assert Exit Code
49     - type: CMD
50       op: ASSERT
51       response: resp2
52       action: EXITCODE
53       expected: 0
54

```

2. Die aufrufende Test-Suite: `test_suite.yaml`

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```

10     # 1. Execute and store full response
11     - type: CALL
12       command: "reusable.demo_reusable"
13       parameters:
14         command: "mvn --version"
15
16   - id: TC_CMD_02
17     description: Demo Reusable basic - verbose
18     stage: '*'
19     priority: HIGH
20     variables:
21       command: "java -version"
22       command2: "java -version"
23
24     steps:
25       # 1. Execute and store full response
26       - type: CALL
27         command: "reusable.demo_reusable"
28         parameters:
29           verbose: "true"
30           command: "mvn --version"
31
32
33   - id: TC_CMD_03
34     description: Demo Reusable basic - R-Buffer vs B-Buffer
35     stage: '*'
36     priority: HIGH
37     variables:
38       command: "java -version"
39       command2: "java -version"
40     steps:
41       # 1. Execute and store full response
42       - type: CALL
43         command: "reusable.demo_reusable"
44         parameters:
45           verbose: "true"
46           command: "mvn -version"
47
48   - id: TC_CMD_04
49     description: Demo Reusable basic - Responses from Reusable Module
50     stage: '*'
51     priority: HIGH
52     variables:
53       command: "java -version"
54       command2: "java -version"
55     steps:
56       # 1. Execute and store full response
57       - type: CALL
58         command: "reusable.demo_reusable"
59         parameters:
60           verbose: "true"
61           command: "mvn --version"
62
63       # 4. Buffer response 1
64       - type: BUFFER
65         op: EXEC
66         name: "RResponse1"
67         action: TEXT
68

```

Wenn wir das zusammen starten, erhalten wir folgende Ausgabe:

```
1 [0;36m>>> YAML loaded in: 88 ms[0m
```

```

2 =====
3 TEST CASE: TC_CMD_01
4 DESC      : Demo Reusable basic
5 STAGE     : * | PRI0: HIGH
6 -----
7 VARIABLES:
8   command2      = java -version
9   command       = java -version
10 -----
11
12 [0;34m>>> DSL[0m      # 1. Execute and store full response
13 [0;34m>>> DSL[0m      - type: CALL
14 [0;34m>>> DSL[0m      command: "reusable.demo_reusable"
15 [0;34m>>> DSL[0m      parameters:
16 [0;34m>>> DSL[0m      command: "mvn --version"
17 [0;32m>>> CALL STEP FINISHED | Status: SUCCESS | Duration: 453 ms[
18
19 -----
20 TEST RESULT: TC_CMD_01                                [[0;32mPASS
21 Time: 0,65s | Steps: 1
22 -----
23
24 =====
25 TEST CASE: TC_CMD_02
26 DESC      : Demo Reusable basic - verbose
27 STAGE     : * | PRI0: HIGH
28 -----
29 VARIABLES:
30   command2      = java -version
31   command       = java -version
32 -----
33
34 [0;34m>>> DSL[0m      # 1. Execute and store full response
35 [0;34m>>> DSL[0m      - type: CALL
36 [0;34m>>> DSL[0m      command: "reusable.demo_reusable"
37 [0;34m>>> DSL[0m      parameters:
38 [0;34m>>> DSL[0m      verbose: "true"
39 [0;34m>>> DSL[0m      command: "mvn --version"
40
41 [0;34m>>> DSL[0m      # 1. Execute and store full response based on R-
42 [0;34m>>> DSL[0m      - type: CMD
43 [0;34m>>> DSL[0m      op: EXEC
44 [0;34m>>> DSL[0m      command: "{B[command]}"
45 [0;34m>>> DSL[0m      response: resp1
46 [0;32m>>> CMD [0m: java -version
47 [0;32m<<< OUT [0m: openjdk version "21.0.10" 2026-01-20 LTS
48 [0;32m<<< OUT [0m: OpenJDK Runtime Environment Temurin-21.0.
49 [0;32m<<< OUT [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
50 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 78 ms[0m
51
52 [0;34m>>> DSL[0m      # 2. Assert Exit Code
53 [0;34m>>> DSL[0m      - type: CMD
54 [0;34m>>> DSL[0m      op: ASSERT
55 [0;34m>>> DSL[0m      response: resp1
56 [0;34m>>> DSL[0m      action: EXITCODE
57 [0;34m>>> DSL[0m      expected: 0
58 [0;32m>>> ASSERT [0m: EXITCODE | Expected: [0]
59 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
60
61 [0;34m>>> DSL[0m      # 2. Execute and store full response based on R-
62 [0;34m>>> DSL[0m      - type: CMD
63 [0;34m>>> DSL[0m      op: EXEC
64 [0;34m>>> DSL[0m      command: "{R[command]}"
65 [0;34m>>> DSL[0m      response: resp1
66 [0;32m>>> CMD [0m: mvn --version
67 [0;32m<<< OUT [0m: Apache Maven 3.9.3 (21122926829f1ead511c9
68 [0;32m<<< OUT [0m: Maven home: C:\Lokale_Programme\apache-ma

```

```

69      [0;32m<<< OUT      [0m: Java version: 21.0.4, vendor: Red Hat, In
70      [0;32m<<< OUT      [0m: Default locale: de_DE, platform encoding:
71      [0;32m<<< OUT      [0m: OS name: "windows 11", version: "10.0", a
72      [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 200 ms[0m
73
74      [0;34m>>> DSL[0m      # 3. Assert Exit Code
75      [0;34m>>> DSL[0m      - type: CMD
76      [0;34m>>> DSL[0m      op: ASSERT
77      [0;34m>>> DSL[0m      response: resp1
78      [0;34m>>> DSL[0m      action: EXITCODE
79      [0;34m>>> DSL[0m      expected: 0
80      [0;32m>>> ASSERT      [0m: EXITCODE | Expected: [0]
81      [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
82
83      [0;34m>>> DSL[0m      # 4. Execute and store full response based on R-
84      [0;34m>>> DSL[0m      - type: CMD
85      [0;34m>>> DSL[0m      op: EXEC
86      [0;34m>>> DSL[0m      command: "{B[command]}"
87      [0;34m>>> DSL[0m      response: resp1
88      [0;32m>>> CMD          [0m: java -version
89      [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
90      [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
91      [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
92      [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 73 ms[0m
93
94      [0;34m>>> DSL[0m      # 5. Assert Exit Code
95      [0;34m>>> DSL[0m      - type: CMD
96      [0;34m>>> DSL[0m      op: ASSERT
97      [0;34m>>> DSL[0m      response: resp1
98      [0;34m>>> DSL[0m      action: EXITCODE
99      [0;34m>>> DSL[0m      expected: 0
100     [0;32m>>> ASSERT      [0m: EXITCODE | Expected: [0]
101     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
102
103     [0;34m>>> DSL[0m      # 6. Execute and store full response based on B-
104     [0;34m>>> DSL[0m      - type: CMD
105     [0;34m>>> DSL[0m      op: EXEC
106     [0;34m>>> DSL[0m      command: "{B[command2]}"
107     [0;34m>>> DSL[0m      response: resp2
108     [0;32m>>> CMD          [0m: java -version
109     [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
110     [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
111     [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
112     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 76 ms[0m
113
114     [0;34m>>> DSL[0m      # 7. Assert Exit Code
115     [0;34m>>> DSL[0m      - type: CMD
116     [0;34m>>> DSL[0m      op: ASSERT
117     [0;34m>>> DSL[0m      response: resp2
118     [0;34m>>> DSL[0m      action: EXITCODE
119     [0;34m>>> DSL[0m      expected: 0
120     [0;32m>>> ASSERT      [0m: EXITCODE | Expected: [0]
121     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
122     [0;32m>>> CALL STEP FINISHED | Status: SUCCESS | Duration: 434 ms[
123
124     -----
125     TEST RESULT: TC_CMD_02                                     [[0;32mPASS
126     Time: 0,44s | Steps: 1
127     -----
128
129     =====
130     TEST CASE: TC_CMD_03
131     DESC      : Demo Reusable basic - R-Buffer vs B-Buffer
132     STAGE     : * | PRI0: HIGH
133     -----
134     VARIABLES:
135     command2  = java -version

```

```

136      command      = java -version
137      -----
138
139 [0;34m>>> DSL[0m      # 1. Execute and store full response
140 [0;34m>>> DSL[0m      - type: CALL
141 [0;34m>>> DSL[0m      command: "reusable.demo_reusable"
142 [0;34m>>> DSL[0m      parameters:
143 [0;34m>>> DSL[0m          verbose: "true"
144 [0;34m>>> DSL[0m          command: "mvn -version"
145
146 [0;34m>>> DSL[0m      # 1. Execute and store full response based on R-
147 [0;34m>>> DSL[0m      - type: CMD
148 [0;34m>>> DSL[0m      op: EXEC
149 [0;34m>>> DSL[0m      command: "{B[command]}"
150 [0;34m>>> DSL[0m      response: resp1
151 [0;32m>>> CMD          [0m: java -version
152 [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
153 [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
154 [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
155 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 76 ms[0m
156
157 [0;34m>>> DSL[0m      # 2. Assert Exit Code
158 [0;34m>>> DSL[0m      - type: CMD
159 [0;34m>>> DSL[0m      op: ASSERT
160 [0;34m>>> DSL[0m      response: resp1
161 [0;34m>>> DSL[0m      action: EXITCODE
162 [0;34m>>> DSL[0m      expected: 0
163 [0;32m>>> ASSERT      [0m: EXITCODE | Expected: [0]
164 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
165
166 [0;34m>>> DSL[0m      # 2. Execute and store full response based on R-
167 [0;34m>>> DSL[0m      - type: CMD
168 [0;34m>>> DSL[0m      op: EXEC
169 [0;34m>>> DSL[0m      command: "{R[command]}"
170 [0;34m>>> DSL[0m      response: resp1
171 [0;32m>>> CMD          [0m: mvn -version
172 [0;32m<<< OUT          [0m: Apache Maven 3.9.3 (21122926829f1lead511c9
173 [0;32m<<< OUT          [0m: Maven home: C:\Lokale_Programme\apache-ma
174 [0;32m<<< OUT          [0m: Java version: 21.0.4, vendor: Red Hat, In
175 [0;32m<<< OUT          [0m: Default locale: de_DE, platform encoding:
176 [0;32m<<< OUT          [0m: OS name: "windows 11", version: "10.0", a
177 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 219 ms[0]
178
179 [0;34m>>> DSL[0m      # 3. Assert Exit Code
180 [0;34m>>> DSL[0m      - type: CMD
181 [0;34m>>> DSL[0m      op: ASSERT
182 [0;34m>>> DSL[0m      response: resp1
183 [0;34m>>> DSL[0m      action: EXITCODE
184 [0;34m>>> DSL[0m      expected: 0
185 [0;32m>>> ASSERT      [0m: EXITCODE | Expected: [0]
186 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
187
188 [0;34m>>> DSL[0m      # 4. Execute and store full response based on R-
189 [0;34m>>> DSL[0m      - type: CMD
190 [0;34m>>> DSL[0m      op: EXEC
191 [0;34m>>> DSL[0m      command: "{B[command]}"
192 [0;34m>>> DSL[0m      response: resp1
193 [0;32m>>> CMD          [0m: java -version
194 [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
195 [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
196 [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
197 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 74 ms[0m
198
199 [0;34m>>> DSL[0m      # 5. Assert Exit Code
200 [0;34m>>> DSL[0m      - type: CMD
201 [0;34m>>> DSL[0m      op: ASSERT
202 [0;34m>>> DSL[0m      response: resp1

```

```

203 [0;34m>>> DSL[0m      action: EXITCODE
204 [0;34m>>> DSL[0m      expected: 0
205     [0;32m>>> ASSERT    [0m: EXITCODE | Expected: [0]
206     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
207
208 [0;34m>>> DSL[0m      # 6. Execute and store full response based on B-
209 [0;34m>>> DSL[0m      - type: CMD
210 [0;34m>>> DSL[0m      op: EXEC
211 [0;34m>>> DSL[0m      command: "{B[command2]}"
212 [0;34m>>> DSL[0m      response: resp2
213     [0;32m>>> CMD        [0m: java -version
214     [0;32m<<< OUT        [0m: openjdk version "21.0.10" 2026-01-20 LTS
215     [0;32m<<< OUT        [0m: OpenJDK Runtime Environment Temurin-21.0.
216     [0;32m<<< OUT        [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
217     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 76 ms[0m
218
219 [0;34m>>> DSL[0m      # 7. Assert Exit Code
220 [0;34m>>> DSL[0m      - type: CMD
221 [0;34m>>> DSL[0m      op: ASSERT
222 [0;34m>>> DSL[0m      response: resp2
223 [0;34m>>> DSL[0m      action: EXITCODE
224 [0;34m>>> DSL[0m      expected: 0
225     [0;32m>>> ASSERT    [0m: EXITCODE | Expected: [0]
226     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
227     [0;32m>>> CALL STEP FINISHED | Status: SUCCESS | Duration: 452 ms[
228
229 -----
230 TEST RESULT: TC_CMD_03                                [[0;32mPASS
231 Time: 0,45s | Steps: 1
232 -----
233
234 =====
235 TEST CASE: TC_CMD_04
236 DESC      : Demo Reusable basic - Responses from Reusable Module
237 STAGE      : * | PRI0: HIGH
238 -----
239 VARIABLES:
240     command2      = java -version
241     command        = java -version
242 -----
243
244 [0;34m>>> DSL[0m      # 1. Execute and store full response
245 [0;34m>>> DSL[0m      - type: CALL
246 [0;34m>>> DSL[0m      command: "reusable.demo_reusable"
247 [0;34m>>> DSL[0m      parameters:
248 [0;34m>>> DSL[0m      verbose: "true"
249 [0;34m>>> DSL[0m      command: "mvn --version"
250
251 [0;34m>>> DSL[0m      # 1. Execute and store full response based on R-
252 [0;34m>>> DSL[0m      - type: CMD
253 [0;34m>>> DSL[0m      op: EXEC
254 [0;34m>>> DSL[0m      command: "{B[command]}"
255 [0;34m>>> DSL[0m      response: resp1
256     [0;32m>>> CMD        [0m: java -version
257     [0;32m<<< OUT        [0m: openjdk version "21.0.10" 2026-01-20 LTS
258     [0;32m<<< OUT        [0m: OpenJDK Runtime Environment Temurin-21.0.
259     [0;32m<<< OUT        [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
260     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 73 ms[0m
261
262 [0;34m>>> DSL[0m      # 2. Assert Exit Code
263 [0;34m>>> DSL[0m      - type: CMD
264 [0;34m>>> DSL[0m      op: ASSERT
265 [0;34m>>> DSL[0m      response: resp1
266 [0;34m>>> DSL[0m      action: EXITCODE
267 [0;34m>>> DSL[0m      expected: 0
268     [0;32m>>> ASSERT    [0m: EXITCODE | Expected: [0]
269     [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m

```

```

270
271 [0;34m>>> DSL[0m      # 2. Execute and store full response based on R-
272 [0;34m>>> DSL[0m      - type: CMD
273 [0;34m>>> DSL[0m      op: EXEC
274 [0;34m>>> DSL[0m      command: "{R[command]}"
275 [0;34m>>> DSL[0m      response: resp1
276 [0;32m>>> CMD          [0m: mvn --version
277 [0;32m<<< OUT          [0m: Apache Maven 3.9.3 (21122926829f1ead511c9
278 [0;32m<<< OUT          [0m: Maven home: C:\Lokale_Programme\apache-ma
279 [0;32m<<< OUT          [0m: Java version: 21.0.4, vendor: Red Hat, In
280 [0;32m<<< OUT          [0m: Default locale: de_DE, platform encoding:
281 [0;32m<<< OUT          [0m: OS name: "windows 11", version: "10.0", a
282 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 205 ms[0m
283
284 [0;34m>>> DSL[0m      # 3. Assert Exit Code
285 [0;34m>>> DSL[0m      - type: CMD
286 [0;34m>>> DSL[0m      op: ASSERT
287 [0;34m>>> DSL[0m      response: resp1
288 [0;34m>>> DSL[0m      action: EXITCODE
289 [0;34m>>> DSL[0m      expected: 0
290 [0;32m>>> ASSERT        [0m: EXITCODE | Expected: [0]
291 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
292
293 [0;34m>>> DSL[0m      # 4. Execute and store full response based on R-
294 [0;34m>>> DSL[0m      - type: CMD
295 [0;34m>>> DSL[0m      op: EXEC
296 [0;34m>>> DSL[0m      command: "{B[command]}"
297 [0;34m>>> DSL[0m      response: resp1
298 [0;32m>>> CMD          [0m: java -version
299 [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
300 [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
301 [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
302 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 75 ms[0m
303
304 [0;34m>>> DSL[0m      # 5. Assert Exit Code
305 [0;34m>>> DSL[0m      - type: CMD
306 [0;34m>>> DSL[0m      op: ASSERT
307 [0;34m>>> DSL[0m      response: resp1
308 [0;34m>>> DSL[0m      action: EXITCODE
309 [0;34m>>> DSL[0m      expected: 0
310 [0;32m>>> ASSERT        [0m: EXITCODE | Expected: [0]
311 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
312
313 [0;34m>>> DSL[0m      # 6. Execute and store full response based on B-
314 [0;34m>>> DSL[0m      - type: CMD
315 [0;34m>>> DSL[0m      op: EXEC
316 [0;34m>>> DSL[0m      command: "{B[command2]}"
317 [0;34m>>> DSL[0m      response: resp2
318 [0;32m>>> CMD          [0m: java -version
319 [0;32m<<< OUT          [0m: openjdk version "21.0.10" 2026-01-20 LTS
320 [0;32m<<< OUT          [0m: OpenJDK Runtime Environment Temurin-21.0.
321 [0;32m<<< OUT          [0m: OpenJDK 64-Bit Server VM Temurin-21.0.10+
322 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 73 ms[0m
323
324 [0;34m>>> DSL[0m      # 7. Assert Exit Code
325 [0;34m>>> DSL[0m      - type: CMD
326 [0;34m>>> DSL[0m      op: ASSERT
327 [0;34m>>> DSL[0m      response: resp2
328 [0;34m>>> DSL[0m      action: EXITCODE
329 [0;34m>>> DSL[0m      expected: 0
330 [0;32m>>> ASSERT        [0m: EXITCODE | Expected: [0]
331 [0;32m>>> CMD STEP FINISHED | Status: SUCCESS | Duration: 0 ms[0m
332 [0;32m>>> CALL STEP FINISHED | Status: SUCCESS | Duration: 432 ms[
333
334 [0;34m>>> DSL[0m      # 4. Buffer response 1
335 [0;34m>>> DSL[0m      - type: BUFFER
336 [0;34m>>> DSL[0m      op: EXEC

```

```

337 [0;34m>>> DSL[0m          name: "RResponse1"
338 [0;34m>>> DSL[0m          action: TEXT
339 [0;32m>>> BUFFER          [0m: [RResponse1] | Store value [null]
340 [0;32m>>> BUFFER STEP FINISHED | Status: SUCCESS | Duration: 0 ms[
341
342 -----
343 TEST RESULT: TC_CMD_04                                     [[0;32mPASS
344 Time: 0,43s | Steps: 2
345 -----
346
347
348 Finalizing suite and cleaning up resources...
349 =====
350                               FINAL EXECUTION SUMMARY
351 =====
352 Total Test Cases : 4
353 Passed           : [0;32m4[0m
354 Failed           : [0m0[0m
355 Success Rate     : 100,0%
356 Total Time       : 1,98s
357 =====
358 OVERALL STATUS   : [[0;32mSUCCESS[0m]
359 =====
360
361 >>> HTML report successfully generated at: target/reports/TestReport.h
362

```

Troubleshooting & Best Practices

• Fehlermeldung `File not found` beim **CALL-Schritt**:

- *Ursache*: Die Notation im Feld `command` stimmt nicht mit der Ordnerstruktur überein. Wenn dort `reusable.demo_reusable` angegeben ist, muss zwingend ein Ordner namens `reusable` direkt neben der testausführenden YAML-Datei existieren, welcher die Datei `demo_reusable.yaml` enthält.

• Variable liefert unerwartet `"NULL"` oder bricht ab:

- *Ursache*: Verwechslung von `{B[...]}` und `{R[...]}`. Denken Sie daran: Möchten Sie Werte nutzen, die im aktuellen Testschritt unter `parameters:` definiert wurden, müssen Sie im Modul zwingend die Syntax `{R[parameter_name]}` wählen. Globale Testfall-Variablen werden immer via `{B[variable_name]}` angesprochen.

• Sichtbarkeit von Responses im Haupttest:

- *Verhalten*: Wie in `TC_CMD_05` gezeigt, sind Responses (wie `response: resp1`), die innerhalb eines Reusable Modules deklariert und befüllt werden, nach Beendigung des `CALL`-Schritts weiterhin im globalen Kontext verfügbar. Sie können im darauffolgenden Haupttest-Schritt problemlos über Operationen wie `BUFFER` oder `ASSERT` weiterverwendet werden.